Chapter 14, Problem 8PP

Problem

A parallel resonant circuit has R = 100 kΩ, L = 50 mH, and C = 2 nF. Calculate $\omega 0$, $\omega 1$, $\omega 2$, Q, and B.

Step-by-step solution

Step 1 of 4

Write the expression for the resonance frequency for a parallel RLC circuit.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute $50\ \text{mH}$ for L and $2\ \text{nF}$ for C.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{(50\times10^{-3})(2\times10^{-9})}}\\ &= \frac{1}{\sqrt{100\times10^{-12}}}\\ &= \frac{1}{10^{-5}}\\ &= 100\ \text{krad/s}\end{aligned}$$

Therefore, the resonant frequency, ω_0 is $\boxed{100\ \text{krad/s}}$.

Step 2 of 4

Write the expression for the Quality factor of a parallel RLC circuit.

$$Q = \frac{R}{\omega_0 L}$$

Substitute $100\ \text{k}\Omega$ for R, $100\ \text{krad/s}$ for ω_0 and $50\ \text{mH}$ for L.

$$\begin{aligned}Q &= \frac{100\times10^{3}}{(100\times10^{3})(50\times10^{-3})}\\ &= \frac{1}{50\times10^{-3}}\\ &= 20\end{aligned}$$

Therefore, the quality factor, Q of the circuit is $\boxed{20}$.

Step 3 of 4

Calculate the Bandwidth of the circuit.

$$B = \frac{\omega_0}{Q}$$

Substitute 100 krad/s for ω_0 and 20 for Q.

$$\begin{aligned} B &= \frac{100 \times 10^3}{20} \\ &= 5000 \text{ rad/s} \\ &= 5 \text{ krad/s} \end{aligned}$$

Therefore, the bandwidth, B of the circuit is $\boxed{5 \text{ krad/s}}$.

Step 4 of 4

Calculate the half power frequency ω_1.

$$\omega_1 = \omega_0 - \frac{B}{2}$$

Substitute 100 krad/s for ω_0 and 5 krad/s for B.

$$\begin{aligned} \omega_1 &= 100 \times 10^3 - \frac{5000}{2} \\ &= 100 \times 10^3 - 2500 \\ &= 97.5 \text{ krad/s} \end{aligned}$$

Calculate the half power frequency ω_2.

$$\omega_2 = \omega_0 + \frac{B}{2}$$

Substitute 100 krad/s for ω_0 and 5 krad/s for B.

$$\begin{aligned} \omega_2 &= 100 \times 10^3 + \frac{5000}{2} \\ &= 100 \times 10^3 + 2500 \\ &= 102.5 \text{ krad/s} \end{aligned}$$

Therefore, the half power frequencies, ω_1 and ω_2 are $\boxed{97.5 \text{ krad/s}}$ and $\boxed{102.5 \text{ krad/s}}$.